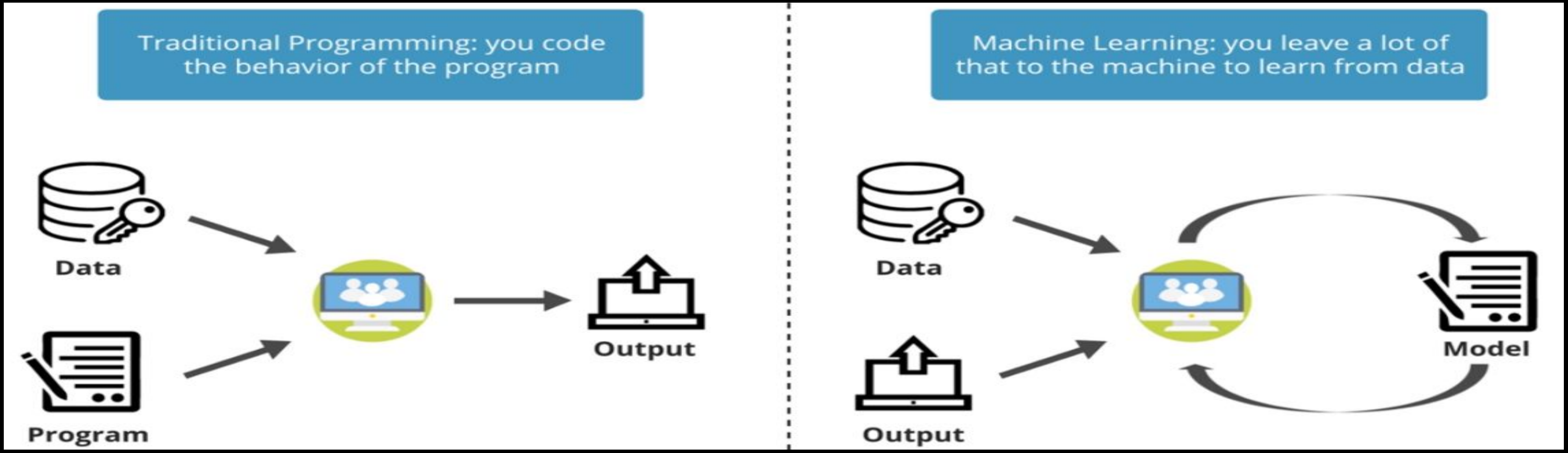
Introduction to Machine Learning

- ✔ Overview – Design of a Learning System
- ✔ Types of Machine Learning
- ✔ Supervised Learning and Unsupervised Learning
- ✔ Mathematical Foundations of Machine Learning
- ✔ Applications of Machine Learning , Machine Learning Foundations

Traditional Programming Vs Machine Learning

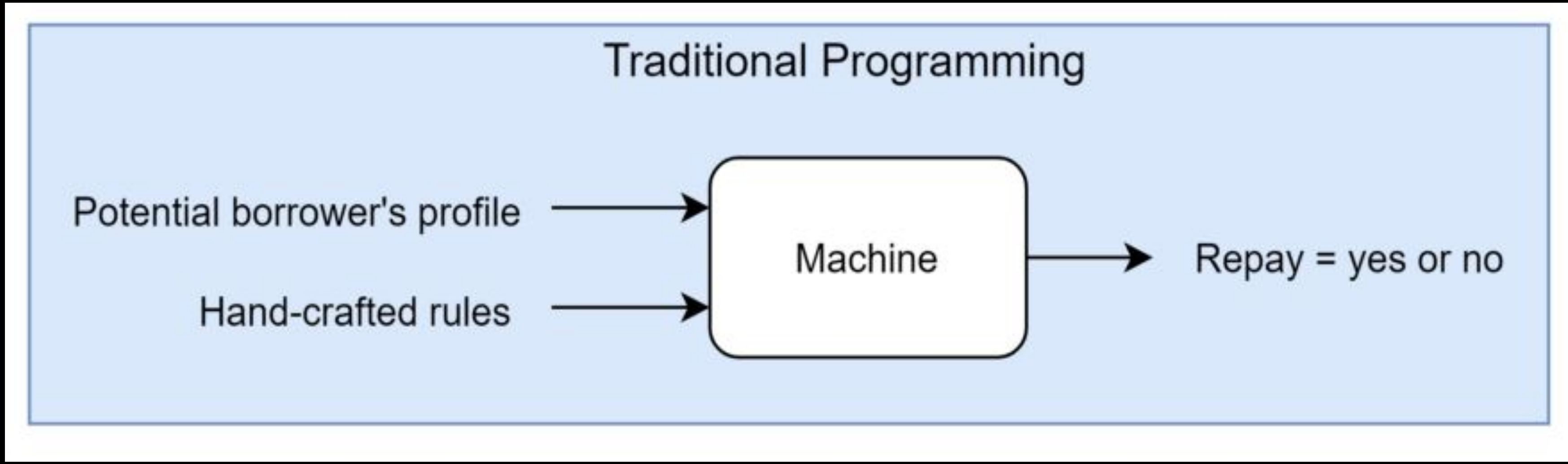
- Traditional Programming refers to any manually created program that uses input data and runs on a computer to produce the output.

- In Machine Learning, also known as augmented analytics, the input data and output are fed to an algorithm to create a program. This yields powerful insights that can be used to predict future outcomes.



Traditional Programming Vs Machine Learning Example

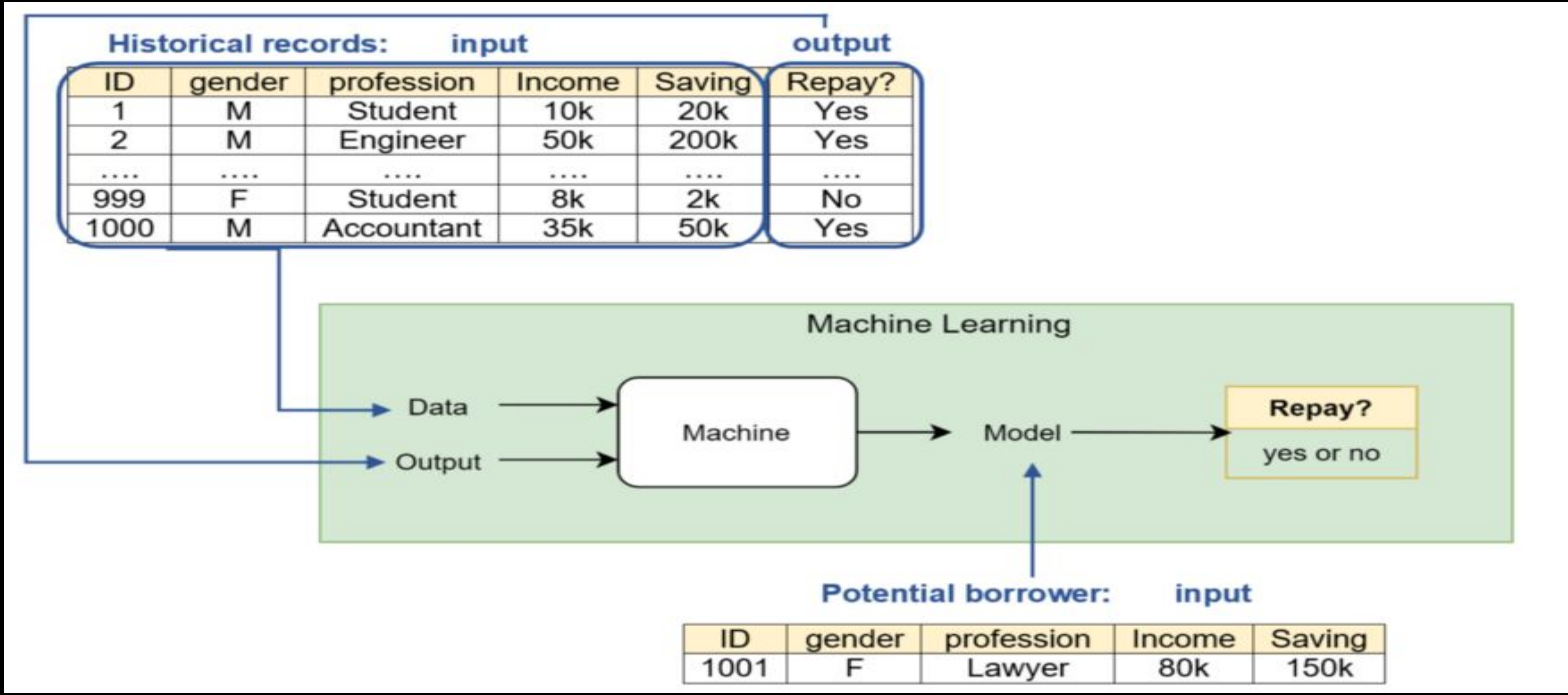
- Credit Risk Estimation: Our goal is to automatically evaluate the risk that a potential borrower will pay a loan or not.
- Traditional Approach



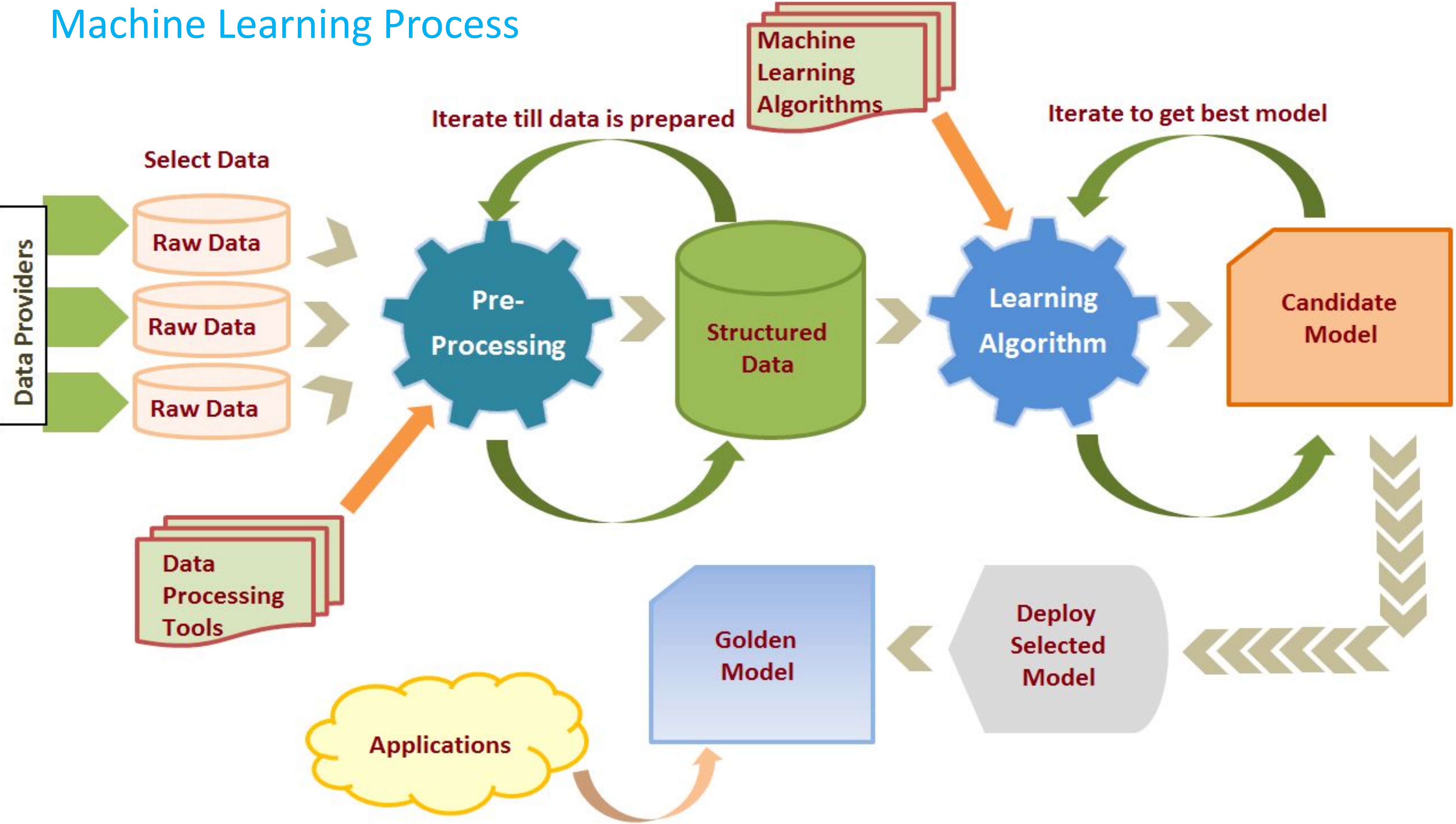
Eg. If income > 150k AND saving > 200k, then Repay = Yes
Eg. If income < 150k AND saving < 200k, then Repay = No

Traditional Programming Vs Machine Learning Example

- Credit Risk Estimation: Our goal is to automatically evaluate the risk that a potential borrower will pay a loan or not.
- ML approach



Machine Learning Process



Machine Learning Process

1.Data Preparation and Data Pre-processing

Data preprocessing is a process of preparing the raw data and making it suitable for a machine learning model.

It includes Data Integration , Data transformation , Data Cleaning and data reduction

2. Model selection

There are many models that researchers and data scientists have created over the years.

Some are very well suited for image data, others for sequences (like text, or music), some for numerical data, others for text-based data.

Model selection in machine learning refers to the process of choosing the best model from a set of candidate models for a specific task or problem

3. Training

In this phase we use data sets to incrementally improve model's ability.

Training data set

- ✔ Is the data set on which model is trained
- ✔ Train data set is data set from which the model has learned the experiences.
- ✔ Training sets are used to fit and tune models.

Test data set

- ✔ Test data is the data that is used to check if the model has learned good enough from the experiences it got in the train data set.
- ✔ Test sets are "unseen" data to evaluate your models.

4. Evaluation

Evaluation allows us to test our model against data that has never been used for training.

5. Model Deployment

Deployment of an ML-model simply means the integration of the finalized model into a production environment and getting results to make business decisions.

Revise

1. Explain steps of machine learning process
2. Define the terms training data and testing data
3. Compare traditional programming and machine learning

Terms used in types of machine learning

1.Labeled Data

Data consisting of a set of training examples, where each example is a pair consisting of an input and a desired output value

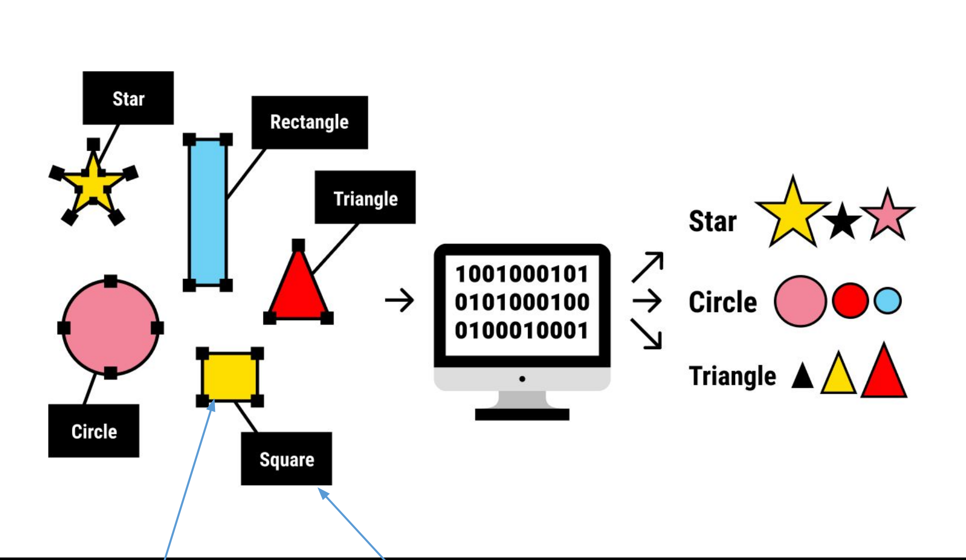
Ex.

1.

Data set shown in the image is labeled

nage is labeled by its

shap



Input

Output

Ex.
2.

Data set shown in the image is labeled
cord is labeled with
nosis.

ere are , Strep Throat,

Patient Id	Sore throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	No	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

Input

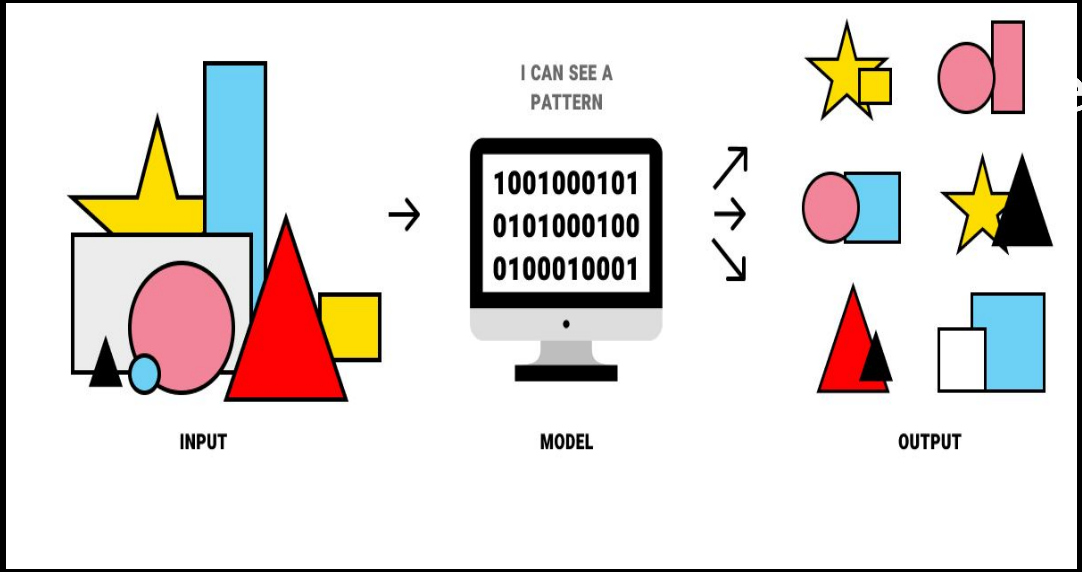
Output

2.Unabled Data

Unlabeled data doesn't have any meaningful tags or labels .

Ex.

1.



Data set shown in the image is

et. shape.

Ex.
2.

Patient Id	Sore throat	Fever	Swollen Glands	Congestion	Headache
1	Yes	Yes	Yes	Yes	Yes
2	No	No	No	Yes	Yes
3	Yes	Yes	No	Yes	No
4	Yes	No	No	No	No
5	No	Yes	No	Yes	No
6	No	No	No	Yes	No
7	No	No	Yes	No	No
8	Yes	No	No	Yes	Yes
9	No	Yes	No	Yes	Yes
10	Yes	Yes	No	Yes	Yes

Input

Data set shown in the image is Unlabeled

Types of machine Learning

1. Supervised learning

- ✔ Supervised learning is the types of machine learning In which machine learning algorithms are feed with the well labeled data.ie. Input and its associated output.

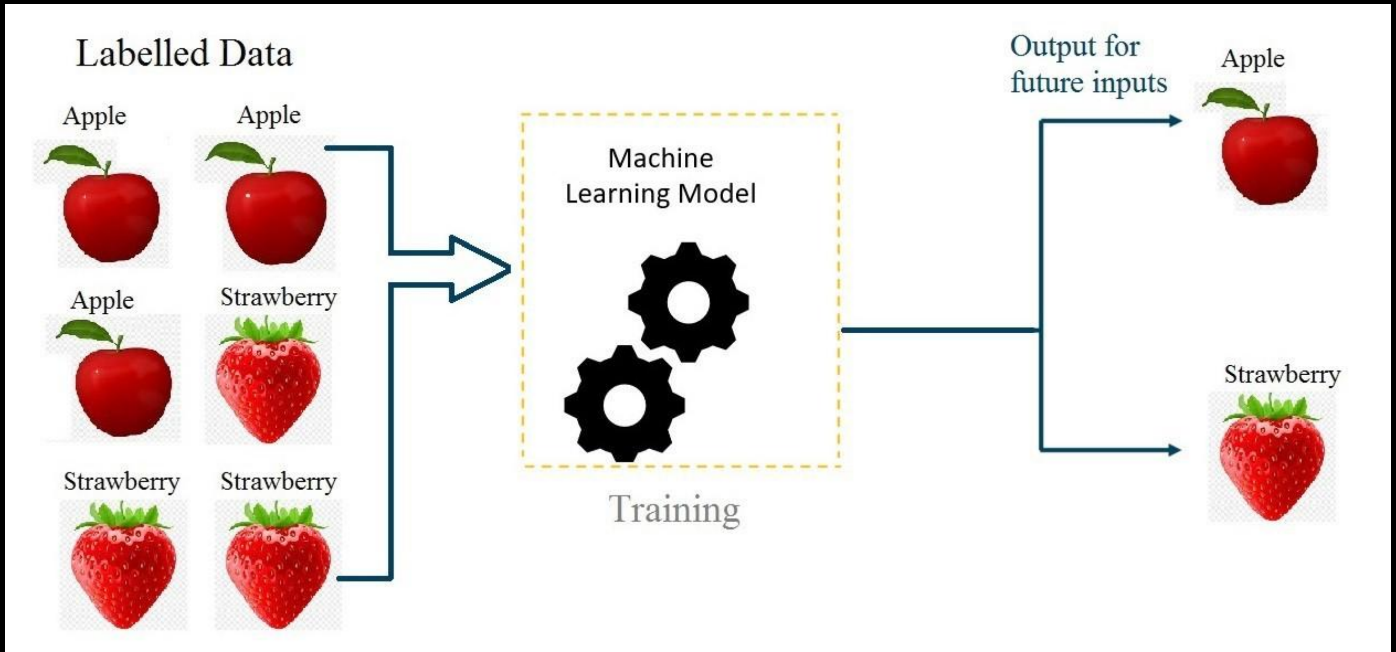
- ✔ From the labeled data algorithms should be able to learn the patterns

- ✔ Supervised learning algorithms try to model relationships and dependencies between the target prediction output and the input features

- ✔ And then we can predict the output values for new data based on those relationships which it learned from the previous data sets.

- ✔ Are the family of algorithms that which are mainly used in classification and Regression

Supervised learning



Types of machine Learning
2. Unsupervised learning

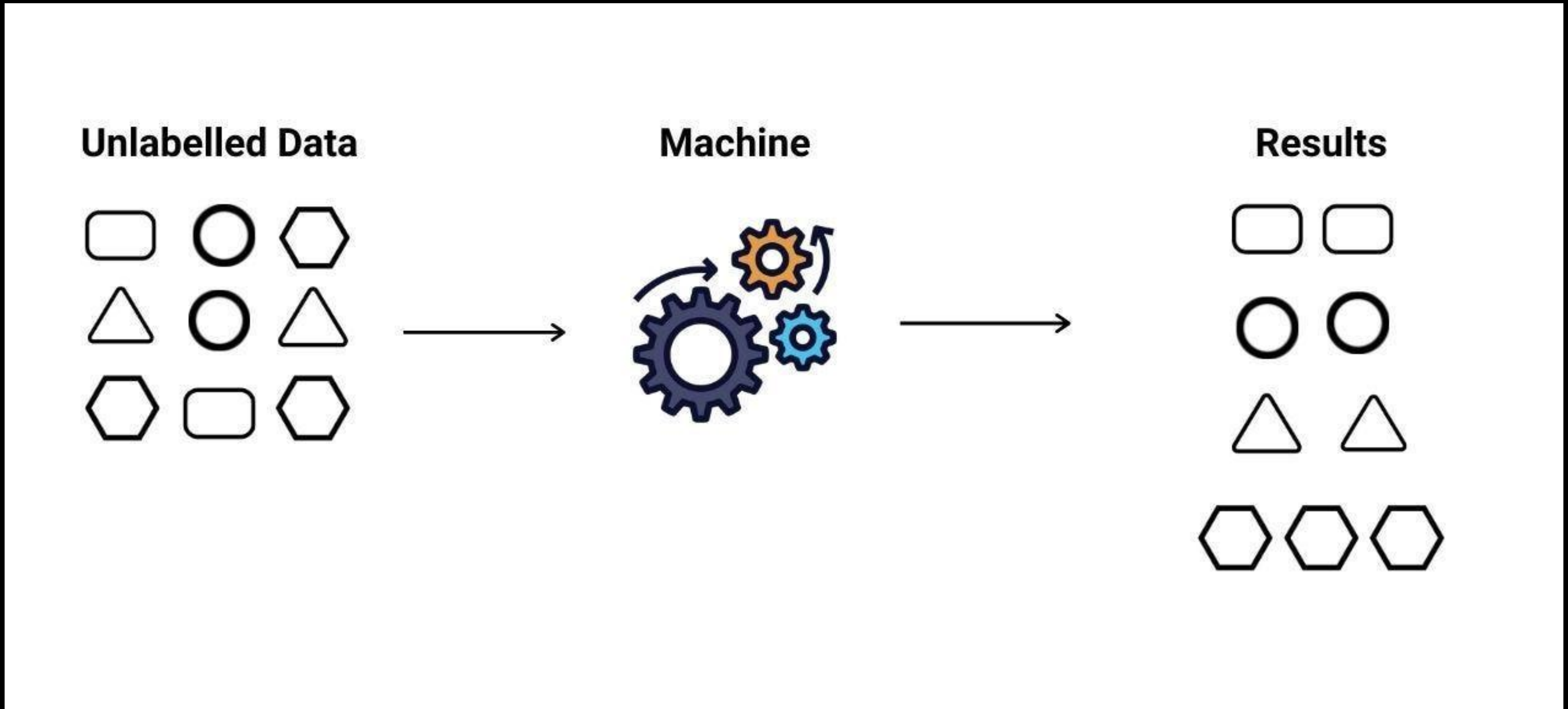
- ✔ Unsupervised learning is a category of machine learning in which we only have the input data to feed to the machine learning algorithm but no corresponding output

- ✔ Algorithm learns the patterns based on the similarity between input data instances .

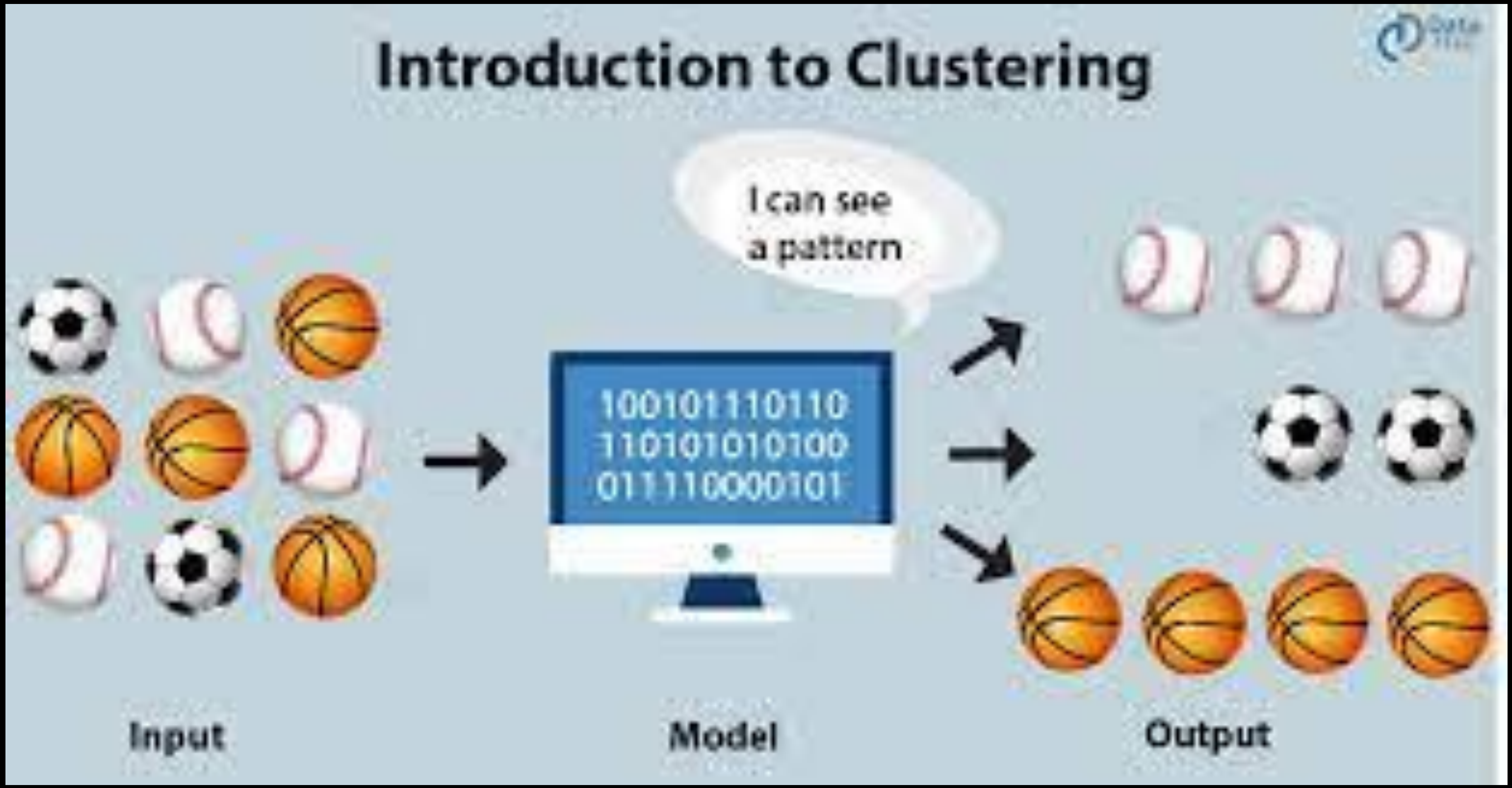
- ✔ Are the family of algorithms that which are mainly used pattern detection and descriptive modeling

- ✔ These algorithms try to use techniques on the input data to mine for rules, detect patterns, and summarize and group the data points which help in deriving meaningful insights and describe the data better to the users.

Unsupervised learning



Unsupervised learning



Revise

1. Explain the terms Labeled data and unlabeled data using suitable example
2. Explain supervised learning using suitable example
3. Explain unsupervised learning suitable example

Machine Learning Foundations

Data Sets

Explanatory variable/independent variable

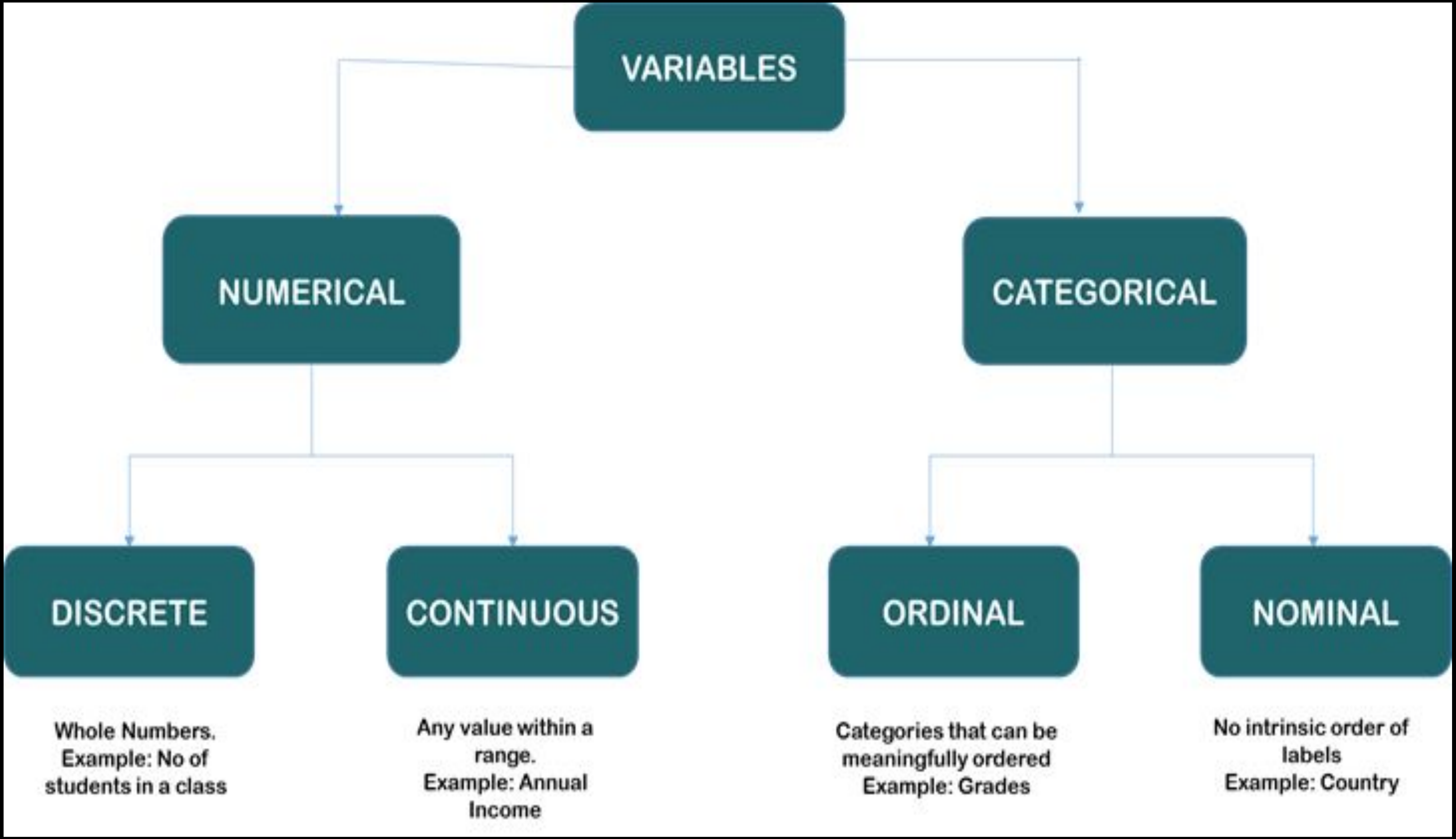
subject identifier

Outcome Variable/dependent variable

one row per experimental subject

	income	happiness
1	3.862647418	2.314488983
2	4.979381382	3.433489759
3	4.923956936	4.599373404
4	3.214372439	2.791113803
5	7.196409251	5.596398273
6	3.729643479	2.458555872
7	4.674517389	3.192991809
8	4.498103821	1.907136833
9		2.942449872
10	4.639914435	3.737941605
11	4.632839514	3.175406147
12	2.773178895	2.009046461
13	7.119478595	5.951814099
14	7.466653196	5.960547308
15		
16	2.55916582	2.898583142
17	2.35479322	1.231167524
18	2.388157248	2.312988055
19	4.755680274	2.666116035
20		2.584729022
21	7.310916026	5.747444104
22	3.528318956	2.546524587

Attribute types



Measures of central tenancy

Mean

- ✔ Centre of data set, can be obtained using a built in aggregation function avg in SQL
- ✔ can be computed using some algebraic function on one or more distributive measures

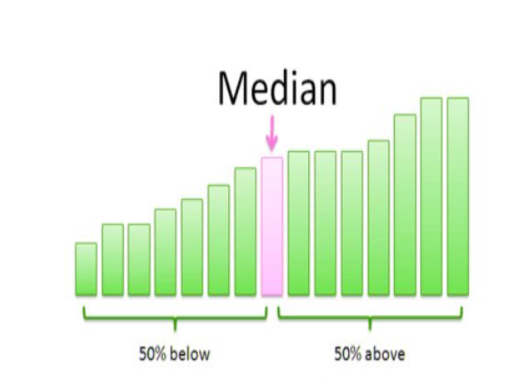
mean() = sum() /count()

- ✔ It is a distributive measure – can be computed parallelly on partitions
- ✔ It is extremely sensitive – Even a small amount of extreme values can corrupt the mean
- ✔ Trimmed mean can be obtained by chopping extreme values

Measures of central tenancy

Median

- ✔ Center of data set, It is the middle value of the ordered set(if n is odd) or average of two middle values(if n is even)
- ✔ It need to be computed on the entire data set as a whole
- ✔ It is not a distributive measure- It cannot be computed by partitioning the data and parallely computing on each partition
- ✔ It is computationally expensive – sorting entire data
- ✔ It is less sensitive to extreme values
- ✔ Ex. (Even number of values) : The median of 3, 3, 5, 9, 11 is 5. If there is an even number of observations, then there is no single middle value; the median is then usually defined to be the mean of the two middle values: so the median of 3, 5, 7, 9 is (5+7)/2 = 6.



Measures of central tenancy

Mode

- ✔ The mode is the value that appears most frequently in a data set.
- ✔ Ex. 3, 3, 6, 9, 16, 16, 16, 27, 27, 37, 48

Mode for above data is 16

- ✔ We can have more than one mode in data. Ex. 3, 3, 3, 9, 16, 16, 16, 27, 37, 48
- ✔ Data sets can be unimodal, bimodal , trimodal etc, Multimodal data sets have two or more modes
- ✔ The mode is not affected by extreme values.

Missing Values

In data pre-processing, it is important to identify and correctly handle the missing values

To deal with missing values one can either

1. Remove the records with missing values
2. Missing values can be replaced by calculated statistical measures i.e. Mean,Median and Mode

What is Feature Scaling ?

- ✔ Machine learning algorithm works on numbers and has no knowledge of what that number represents.

- ✔ It is a method to standardize the independent variables of a dataset within a specific range.

- ✔ Feature Scaling is one of the important pre-processing that is required or standardizing/normalization of the input data.

- ✔ When the range of values are very distinct in each column, we need to scale them to the common level.

Standardization & Normalization

new value

original value

$$X' = \frac{x - \text{mean}(x)}{a}$$

mean

Standard deviation

Standard Deviation

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation

N = the size of the population

x_i = each value from the population

μ = the population mean

new value

original value

$$X' = \frac{x\text{-min}(x)}{\max(x)\text{-min }(x)}$$

Encoding Categorical Data

✔ Most of the Machine learning algorithms can not handle categorical variables unless we convert them to numerical values.

✔ There are many ways we can encode these categorical variables as numbers and use them in an algorithm.

Ex. 1. Label/Ordinal Encoding

2. One Hot Encoding

Encoding Categorical Data

Label/Ordinal Encoding

- ✔ This type of encoding is used when the variables in the data are ordinal.
- ✔ Label encoding converts each label into integer values and the encoded data represents the sequence of labels.

Ex.

Height
Tall
Medium
Short

→

Height
0
1
2

Encoding Categorical Data

One Hot Encoding

- ✔ In One-Hot Encoding, each category of any categorical variable gets a new variable.
- ✔ It maps each category with binary numbers (0 or 1).
- ✔ This type of encoding is used when the data is nominal. Newly created binary features can be considered dummy variables.
- ✔ After one hot encoding, the number of dummy variables depends on the number of categories presented in the data.

Ex.

Index	Animal
0	Dog
1	Cat
2	Sheep
3	Horse
4	Lion

One-Hot code

Index	Dog	Cat	Sheep	Lion	Horse
0	1	0	0	0	0
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	0	1
4	0	0	0	1	0

Splitting the dataset into training dataset and testing dataset

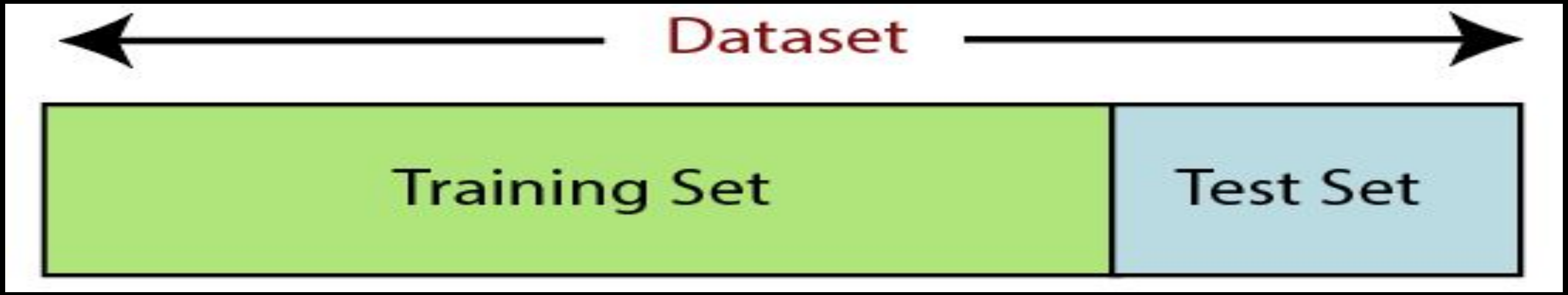
The train-test split is a technique for evaluating the performance of a machine learning algorithm.

It can be used for classification or regression problems and can be used for any supervised learning algorithm.

The procedure involves taking a dataset and dividing it into two subsets.

Train Dataset: Used to fit the machine learning model.

Test Dataset: Used to evaluate the fit machine learning model.



Applications of machine Learning

1. Image Recognition

Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital image

2. Speech Recognition

Speech recognition is a process of converting voice instructions into text, and it is also known as "Speech to text", or "Computer speech recognition."

3. Product recommendations

Machine learning is widely used by various e-commerce and entertainment companies such as Amazon, Netflix, etc., for product recommendation to the user.

Applications of machine Learning

4. Self-driving cars

It is using unsupervised learning method to train the car models to detect people and objects while driving.

5. Email Spam and Malware Filtering

Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning.

6. Medical Diagnosis

In medical science, machine learning is used for diseases diagnoses.

7.online Fraud Detection

Machine learning is making our online transaction safe and secure by detecting fraud transaction.